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Prediction research of cervical cancer clinical events based on recurrent neural network

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Abstract

Through the research on the existing time series prediction technology, most researchers mainly make predictions based on clinical events, without thinking about whether the previous clinical process is standard or not. This paper proposes a two-stage prediction model, RNN-2-DT, based on word vector representation and integrated into the standardized judgment of the diagnosis and treatment process. The model mines standardized clinical mode which is a standard of clinical process, using the method of binary K-means. Meantime, the clinical events prediction model using gated recurrent units (GRU) based on recurrent neural network (RNN) is constructed. The experimental results indicate that, compared with the clinical processes no considering standardized judgments, our model's recall rate and mean average precision are increased by 7.2% and 4.3%, respectively.

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1. Introduction

Cervical cancer is one of the common gynecological malignancies¹. The diagnosis and treatment process is complicated. The clinical decision on diagnosis and treatment of different stages requires consideration of various information, including fertility, lymph node metastasis and other factors. Doctors' diagnosis and treatment decisions

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are often based on experience or search for similar medical records in computer to establish correlation between patient's current condition and historical medical records. And then they make decisions through comparative analysis and guidance of cervical cancer clinical specifications. Obviously, decision-making process is affected by doctor's medical level and is highly subjective. It is difficult to treat according to the clinical path prescribed diagnosis and treatment mode under complex and changeable symptoms.

With the accumulation of large amounts of medical data, data-driven prediction tasks in medical field have aroused extensive attention. Some researchers apply deep learning algorithms to prediction tasks of medical field. Pham et al.² established DeepCare prediction model based on LSTM which transferred the records of diagnosis, examination items and medicine into the cells of LSTM in different forms to predict the patient's diagnosis, clinical items, and whether to re-admission after discharge. Choi et al.³ established Doctor AI model based on GRU. The model converted the diagnosis results, medications into vectors. And then input them into two-layer neural network model to predict clinical items, medication and visiting time of patients' next visit according to the representation of patients at each time.

However, most of clinical prediction tasks are only to predict clinical events of next stage based on historical records, and do not consider whether conform to standardized diagnosis and treatment mode of clinical path. The study⁴ founds that after implementing clinical pathways for patients with cervical cancer, the work efficiency of doctors was improved and problems occurred in diagnosis and treatment process could be prevented in a purposeful manner. Based on this, we mine standardized diagnosis and treatment modes and build a clinical event prediction model based on RNN. The main contributions of this paper include following three aspects:

1) The hospitalization data has the characteristic of semi-sequential. The clinical records of the same day are not distinguished in order, but only the order of treatment records of different dates is considered. This study adopts the method of average pooling to process clinical records of the same day.

2) The time distribution of treatment records during hospitalization is uneven. In this study, time interval was integrated into treatment records. The experimental results showed that the performance of the model was effectively improved after integrating time interval.

3) A two-stage prediction model includes the process of standardized judgment of diagnosis and treatment mode and the process of clinical events prediction. Before predicting future clinical events, we judge whether the previous diagnosis and treatment process is standardized or not. Therefore, the goal of standardizing doctor's diagnosis and treatment behavior has been reached.

2. Embedded vector representation of clinical diagnosis and treatment records

The electronic medical record (EMR) contains a large amount of data. In this study, we extracted four parts of EMR: diagnosis record, medication, doctor's advice and laboratory inspection items. But the data we extracted cannot be used for mining standardized diagnosis and treatment models and the task of predicting clinical events.

In recent years, some researchers have applied word embedding models to the representation of medical concepts, transforming heterogeneous clinical records into structural data. Choi et al.⁵ used Skip-gram algorithm to learn low-dimensional vector representation of medical insurance data of health insurance companies. Zhong et al.⁶ learned patient state representation from patient diagnosis, treatment process and medication by the method of Doc2vec. Chowdhury et al.⁷ constructed a Med2Meta model to obtain a collective representation of all clinical records.

In this study, the method we used is similar to the method used by Choi et al.⁵. However, the difference is that we use the data in the electronic medical record instead of the medical insurance data of the health insurance company, And for the medical records of fixed time intervals, we adopt the method of average pooling instead of random order.

2.1. Definition

Definition 1 Treatment day. During hospitalization, the date includes clinical records.

Definition 2 Clinical events. Clinical events refer to the medication, doctor's advice, and examination items during hospitalization. Several clinical events contained in a treatment day of the patients constitute a collection. Use v_i to represent the i -th clinical events. $e_t = \{v_1, v_2, v_3, \dots, v_n\}$ is the set of clinical events the t -th treatment day.

Definition 3 Clinical events sequence. During the patient's hospitalization, several sets of treatment day constitute the clinical event sequence. $E = \{e_1, e_2, e_3, \dots, e_T\}$ represents a complete hospitalization process.

In this paper, we analyze the history of diagnosis and treatment of patients with cervical cancer longitudinally and express the clinical records of patients as a sequence of clinical events.

2.2. Vector representation model based on Skip-gram

The embedded vector representation model of medical records based on Skip-gram^{8,9} is shown in Figure 1. Because of the characteristics of semi-sequential of clinical records, clinical events that occur on the same treatment day were pooled by average, as shown in equation (1), and are spliced with the diagnosis records to obtain the vector of treatment day which is denoted by e'_t .

$$e_t = \frac{1}{n} \sum_{i=1}^n v_i \quad (1)$$

In equation (1), n is the number of clinical events in a treatment day, and v_i represents the i -th treatment item.

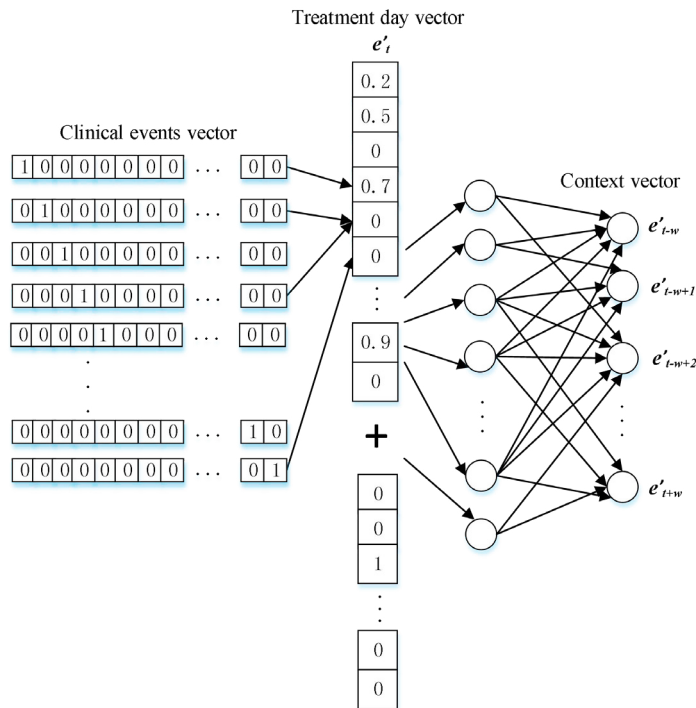


Fig. 1. Embedded vector representation model

The goal of the Skip-gram model is to maximize the logarithm likelihood function, as shown in equation (2).

$$L = \frac{1}{T} \sum_{t=1}^T \sum_{-w \leq j \leq w, j \neq 0} \log p(e'_{t+j} | e'_t) \quad (2)$$

In equation (2), T is the length of treatment record sequence and w is the size of the context window. e'_t is the target treatment day and e'_{t+j} is the context vector. The conditional probability $p(e'_{t+j} | e'_t)$ is defined by the softmax function, as shown in equation (3).

$$p(e'_{t+j}|e'_t) = \frac{\exp(V(e'_{t+j})^T V(e'_t))}{\sum_{i=1}^N \exp(V(e'_i)^T V(e'_t))} \quad (3)$$

In equation (3), $V(e'_i)$ represents the embedded vector representation of e'_i and N represents the total number of clinical events.

In this paper, in order to improve the training speed of the model, the predicted contextual treatment items were transformed into binary classification tasks which predicted whether the vector of contextual treatment day of the target treatment day exists or not. The goal was to obtain the optimal solution of the following objective function.

$$J = \sum_{t=1}^T \left[\log \sigma \left(V(e'_{t+j})^T V(e'_t) \right) + \sum_{i=1}^N \log \sigma \left(V(e'_i)^T V(e'_t) \right) \right] \quad (4)$$

In equation (4), $\sigma(\cdot)$ represents sigmoid function.

$$\sigma(x) = \frac{1}{1+e^{-x}} \quad (5)$$

3. Standardized diagnosis and treatment pattern mining model

In the process of diagnosis and treatment of cervical cancer patients, there will be abnormal behaviors, such as unreasonable medication, that do not conform to diagnosis and treatment mode stipulated by clinical pathway. In order to standardize treatment behaviors and improve the quality of medical treatment, some researchers mines standardized models from a large amount of medical data to find abnormal treatment behaviors of doctors. Xu Xiao et al.¹⁰ proposed an optimized topic model to mine high-quality medical models from medical data. In terms of anomaly detection, Ahmed et al.¹¹ constructed a financial risk detection model based on a clustering algorithm to detect fraud in the financial field. Doshi et al.¹² proposed an efficient hierarchical clustering method to detect content on social networks to identify abnormal content released by users.

In this study, we use the bisecting K-means clustering algorithm based on cosine similarity to mine the standardized diagnosis and treatment patterns in the process of treatment of cervical cancer.

3.1. Bisecting K-means algorithm

The bisecting K-means algorithm is an unsupervised process of dividing the clinical events sequence into multiple diagnosis and treatment modes composed of similar clinical record sequences.

Let $X = \{E_1, E_2, E_3, \dots, E_M\}$ be the sequence of clinical records in the data set. Initially, the number of diagnosis and treatment modes is K . All clinical events sequences are initialized as a cluster and joined cluster table. And then cluster C_i is selected from cluster table to perform binary clustering using K-means method. Finally, we choose cluster with the smallest sum of squared errors (SSE) to join. Repeat the above two steps until K clusters are generated.

The sum of squared errors (SSE) expression is as follows:

$$SSE = \sum_{i=1}^k \sum_{E \in C_i} dist(c_i, E) \quad (6)$$

In equation (6), k is the number of current clusters. E is a sample in cluster C_i and c_i represents the center of cluster C_i . And $dist(c_i, E)$ denotes the distance between the center of cluster C_i and sample E . The expression is as follows:

$$dist(c_i, E) = \frac{1}{\sum_{j=1}^T |\cos \langle c_i, E \rangle [j][j]|} \quad (7)$$

In equation (7), $\cos \langle c_i, E \rangle [j][j]$ denotes the j -th row and j -th column of cosine similarity result matrix. And it is the cosine similarity of the j -th treatment day. T is the length of clinical record sequence, that is, the total number of days for diagnosis and treatment. The cosine similarity function is as follows:

$$\cos \langle c_i, E \rangle = \frac{c_i \times E^T}{\sqrt{(c_i \odot c_i \times P) \times (E \odot E \times P)^T}} \quad (8)$$

In equation (8), P is all “1” column vector with the same dimension as the sample E . And $\odot$ represents the Hadamard product of the matrix and $\times$ represents the matrix multiplication.

At last, the central values of K clusters are selected as the standardized diagnosis and treatment mode sequence.

3.2. Abnormal diagnosis and treatment behavior detection based on cosine similarity

In the detection of abnormal diagnosis and treatment behavior, the clinical record sequence has the characteristics of semi-sequential. Therefore, the cosine similarity is used to measure the similarity between two clinical record sequences on different treatment days.

$$\cos \langle D, c \rangle = \frac{E \times c^T}{\sqrt{(D \odot D \times P) \times (c \odot c \times P)^T}} \quad (9)$$

In equation (9), D represents the sequence of clinical records to be tested and c represents the sequence of standardized diagnosis and treatment mode. P is all “1” column vector with the same dimension as D . And $\odot$ represents the Hadamard product of the matrix and $\times$ represents the matrix multiplication.

4. Prediction model of clinical events

4.1. Recurrent Neural Network(RNN)

In this study, we used GRU¹³ based on RNN¹⁴ to predict clinical events of next treatment day. GRU performs well in sequence-based tasks, ensuring that historical information will not be lost after a long time of propagation.

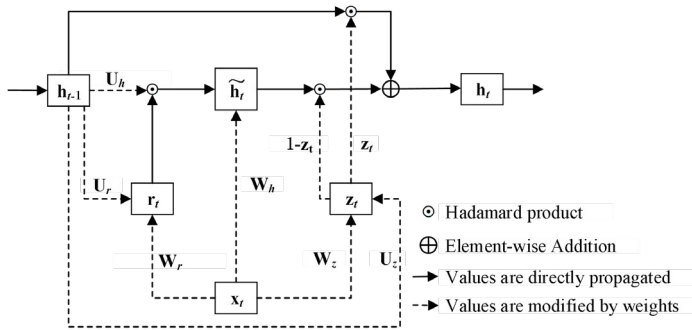


Fig. 2. Gated Recurrent Units (GRU)

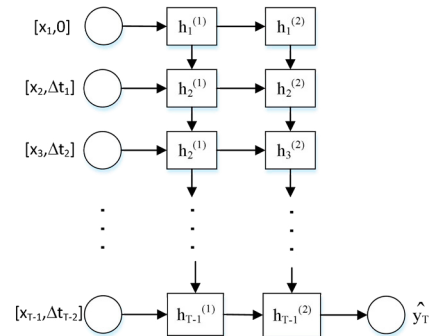


Fig. 3. Prediction model of clinical events

The structure of GRU is shown in Figure 2. The retention and update of historical information are controlled by two gates, reset gate r_t and update gate z_t . Therefore, GRU can control the preservation, reset and update of historical information effectively. Due to the huge amount of data in clinical records, it is difficult for doctors to make decisions about future treatments. This feature of GRU provides a great help to doctors.

The mathematical expression of the GRU model¹⁴ in the original text is as follows:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (10)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (11)$$

$$\tilde{h}_t = \tanh(W_h x_t + r_t \odot U_h h_{t-1} + b_h) \quad (12)$$

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t \quad (13)$$

In the above equation, z_t , r_t are the update gate and the reset gate, respectively. $\tilde{h}_t$ is intermediate memory unit. h_t is the state of the hidden layer. x_t is the input vector, namely $V(e'_t)$. W, U represent the weight matrix and b is the bias. $\odot$ denotes the Hadamard product. $\sigma(\cdot)$ denotes the sigmoid function and $\tanh(\cdot)$ denotes the tanh function.

4.2. Prediction model of clinical events based on GRU

The future clinical events are determined by diagnosis results and treatment items of historical treatment days. In the date set, time interval between treatment days is inconsistent, and the longer interval, the less impact on subsequent treatment items. Therefore, the time interval is an important feature for predicting future clinical events. In this study, time interval of adjacent treatment days and the vector of treatment day were spliced.

The prediction model of clinical events is shown in Figure 3. The spliced vector is used as the input of the model. The patients' disease state representation is learned through the two-layer GRU network. And we use softmax function to predict the result $\hat{y}_T$ which expression is as follows:

$$\hat{y}_T = \text{softmax}(W_s h_{T-1}^{(2)} + b_s) \quad (14)$$

In equation (14), $h_{T-1}^{(2)}$ represents the patient's state representation vector learned through the two-layer GRU neural network. W_s is the weight matrix and b_s is the bias.

The cross entropy between the real result y_{T+1} and the predicted result $\hat{y}_{T+1}$ is used as the loss function:

$$\text{Loss} = \sum_{t=1}^{T-1} (y_{t+1} \log \hat{y}_{t+1} + (1 - y_{t+1}) \log (1 - \hat{y}_{t+1})) \quad (15)$$

5. Experiment and analysis

The flow of the prediction method of cervical cancer clinical events is shown in Figure 4. In this study, we designed four models, which were compared with three baseline models in unstandardized judgment and standardized judgment.

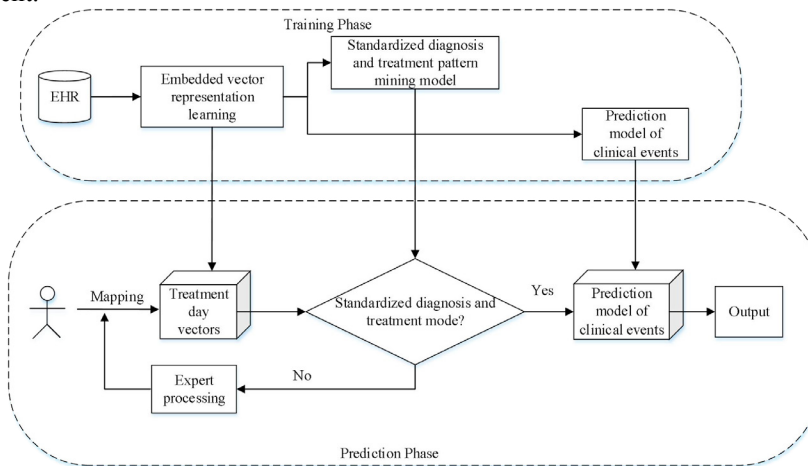


Fig. 4. Clinical events prediction process of cervical cancer

The model is set up as follows:

- 1) RNN-1: Taking treatment day vector as input, there is a hidden layer of neural network based on GRU
- 2) RNN-2: Taking treatment day vector as input, there are two layers of neural network based on GRU.
- 3) RNN-1-DT: Taking the vector of treatment day and the time interval between adjacent treatment days as input, there is a layer of neural network based on GRU.
- 4) RNN-2-DT: Taking the vector of treatment day and the time interval between adjacent treatment days as input, it has two layers of neural networks based on GRU.
- 5) DeepCare²: The diagnostic records, laboratory examination items and medications in the medical time series data are transferred into the cells of LSTM in different forms to construct a DeepCare model based on LSTM.
- 6) FT-LSTM¹⁵: Based on basic LSTM, time-controlled gates are added to LSTM cells.
- 7) HE-LSTM¹⁶: Based on classic LSTM, an event gate is added to LSTM cells to control the renewal state of cells.

5.1. Data set and Experimental setup

The data set is derived from clinical records of 13,000 cervical cancer patients in the electronic medical record database of a hospital from April 2013 to August 2019. It includes the patient's diagnosis records, medications, doctor orders, and laboratory examination items with a total of 518,000 pieces of data. After deleting the patients with less than 2 medical records, there are 11,990 patients with a total of 516,039 pieces of data, with an average of 43 hospital admissions per person. The statistics are shown in Table 1 and Figure 5.

In the training phrase of the model, we take 70% of the data as the training set, 20% as the test set and 10% as the validation set. The results of validation set are used to adjust the hyper-parameters of model. In order to avoid over-fitting of model, dropout is used between layers. When the dropout is 22%, the effect of model is better. The optimizer is Adam. Initial learning rate is 0.01. Weight attenuation coefficient is 0.0001. In order to enhance the expression ability of the model, the number of neurons in the hidden layer is set to 2000.

Table 1. Statistics of the dataset.

Category	Count
# of patients	11,990
# of clinical records	516,039
Avg. # of records per patient	43

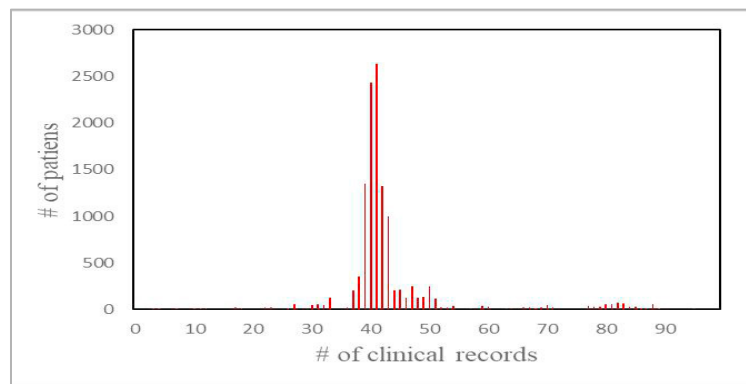


Fig. 5. Statistics of patient clinical records

5.2. Evaluation and analysis of experimental results

In this study, we use Recall@K and MAP@K as the evaluation criteria.

$$\text{Recall@K} = \frac{m'}{n} \quad (16)$$

In equation (16), n' is the number of patients' clinical events. And m' indicates that there are m' correct treatment items in the first K records of the predicted result. Recall@K imitates the behavior of doctors using the most likely treatment items. Therefore, the higher the value of Recall@K, the better the performance of the model. During the experiment, we evaluated the performance of different models under $K=10$, $K=20$ and $K=30$.

$$P(r) = \frac{m'}{r} \quad (17)$$

$$\text{MAP} = \frac{1}{n} \sum_{p=1}^P \sum_{r=1}^{N'} \frac{P(r)}{r} \quad (18)$$

In equation (17) and equation (18), r is the level. n' is the number of patient treatment items. P is the number of patients, and N' is the number of samples. MAP@K (Mean Average Precision) reflects the prediction performance of the model. If the correct treatment items predicted by the model are more advanced, the higher the value of MAP. During the experiment, we evaluated the prediction performance of different models under $K=10$, $K=20$ and $K=30$.

The experimental results are shown in Table 2 and Table 3.

Table 2. Comparison of model Recall rates under different K values.

Model	unstandardized judgment			standardized judgment		
	Recall@K			Recall@K		
	K = 10	K = 20	K = 30	K = 10	K = 20	K = 30
DeepCare	77.1%	78.5%	84.9%	85.4%	87.7%	89.2%
FT-LSTM	75.8%	78.3%	85.2%	83.7%	86.9%	89.9%
HE-LSTM	76.1%	77.3%	80.1%	82.4%	85.8%	87.6%
RNN-1	63.9%	66.1%	70.9%	69.6%	71.1%	73.5%
RNN-2	72.8%	74.2%	81.4%	81.1%	84.8%	86.7%
RNN-1-DT	69.5%	73.2%	80.1%	79.5%	82.3%	85.6%
RNN-2-DT	77.4%	79.3%	85.6%	86.5%	88.7%	90.3%

Table 3. Comparison of model MAP values under different K values.

Model	unstandardized judgment			standardized judgment		
	MAP@K			MAP@K		
	K = 10	K = 20	K = 30	K = 10	K = 20	K = 30
DeepCare	57.3%	59.2%	62.3%	60.5%	63.9%	68.6%
FT-LSTM	56.9%	58.7%	64.3%	60.9%	64.7%	69.1%
HE-LSTM	58.1%	60.0%	65.2%	61.1%	65.3%	69.2%
RNN-1	51.6%	57.8%	60.1%	55.6%	58.9%	63.8%
RNN-2	53.2%	59.7%	63.5%	58.3%	60.7%	65.4%
RNN-1-DT	52.9%	58.6%	62.3%	57.8%	63.2%	68.9%
RNN-2-DT	58.5%	60.1%	65.6%	61.3%	68.5%	70.1%

Compared with the case of unstandardized judgments, Recall@K and MAP@K of our model are increased by 7.2% and 4.3%, respectively. Obviously, our method is much better after standardized judgments. Moreover, Recall@K and MAP@K, in turn, promote 6.8% and 3.2% by adding to time intervals. It can be seen that the model with multiple hidden layers has stronger expressive ability, further revealing the RNN-2-DT has the best prediction effect in case of standardized judgment. The proposed model's Recall@K rises up 1.75% while MAP@K also improves 1.5%, which is dramatically superior to the basic model.

6. Conclusion

This paper proposes a two-stage clinical events prediction model. In the first stage, the standardized diagnosis and treatment mode is mined by bisecting K-means method, used as a standard to judge whether the clinical events sequence to be tested is standard. In the second, we apply GRU into neural network to construct a prediction model. Due to the semi-sequential characteristics of hospitalization data, the treatment items on the same treatment day were pooled on average. Then, skip-gram is used to map hospitalization data to vector space in order to obtain treatment day vector, representing the potential correlation between different treatment days. In terms of data, we use treatment day vectors to mine standardized diagnosis and treatment mode while we train clinical events prediction model through treatment day vector stitched with time interval of the adjacent treatment day. The results indicate that RNN-2-DT has a brilliant prediction effect compared with the traditional models that predict clinical events directly.

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